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(FUZZY LOGIC PROJECT)

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PROJECT TITLE: DIABETES

DATASET NAME: Diabetes Dataset

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1. Dataset Information

Due to unhealthy lifestyles, poor eating habits, and family histories, the incidence of diabetes mellitus, especially type 2 diabetes, has grown to be a global public health problem. Reducing long-term problems like blindness, cardiovascular disease, or kidney failure requires an accurate early diagnosis. Despite their growing popularity, traditional binary classification techniques like logistic regression and decision trees may not be sufficiently versatile to handle scenarios that are problematic or ambiguous. The dataset can be retrieved from the Kaggle website: <https://www.kaggle.com/datasets/akshaydattatraykhare/diabetes-dataset>.

1.1 Objective

This project's main goal is to use the Python programming language to create a fuzzy logic system for diabetes categorisation.

Specific goals consist of:

- To develop a comprehensive fuzzy logic classification system with Kaggle's Pima Indian Diabetes Dataset.
- To create a fuzzy logic framework with fuzzy rules, input and output membership functions, and data preprocessing.
- To use intelligent categorisation that imitates the language thinking processes of medical professionals (e.g., "high glucose", "moderate risk").
- To use a suggested accuracy formula, determine and contrast the classification accuracy of the original dataset classifications with fuzzy logic findings.
- To show how well fuzzy logic handles the overlapping input areas and ambiguity that are common in medical diagnostic data.
- To provide a useful diagnostic tool that can classify the risk levels of diabetes in women of Pima Indian heritage who are at least 21 years old.
- To verify the system's functionality by thorough testing and comparing its correctness to the initial dataset classifications.

1.2 Attribute (Input and Output)

Input

No.	Attribute	Description	Type
1	Pregnancies	Number of times the patient has been pregnant	Integer
2	Glucose	Plasma glucose concentration a few hours after glucose intake	Integer
3	BloodPressure	Diastolic blood pressure (in mm Hg)	Integer
4	SkinThickness	Thickness of the triceps skin fold (in mm), a proxy for body fat	Integer
5	Insulin	Serum insulin level measured two hours after glucose ingestion ($\mu\text{U/mL}$)	Integer
6	BMI	Body Mass Index calculated as weight (kg) / height ² (m ²)	Float
7	DiabetesPedigree Function	A function that scores the likelihood of diabetes based on family history and other factors	Float
8	Age	Age of the patient in years	Integer

Output

No.	Attribute	Description	Type
1	Outcome	Class variable: 0 = non-diabetic, 1 = diabetic	Categorical

1.3 Total Dataset

This dataset contains 768 instances (patients) and 9 attributes. Among these, 8 are input variables and 1 is the output classification variable (Outcome), which serves as the target label.

In a fuzzy system, attributes are typically mapped to linguistic terms such as low, medium, and high. Here's a breakdown of why each attribute is considered significant and suitable for fuzzy membership design:

- **Pregnancies:** While not a direct indicator of diabetes, frequent pregnancies can indicate hormonal changes that influence insulin resistance.
- **Glucose:** A primary indicator of diabetes, especially fasting or postprandial glucose levels.
- **BloodPressure:** High blood pressure often co-occurs with metabolic disorders and increases risk.
- **SkinThickness:** While less predictive in isolation, it's useful as a proxy for subcutaneous fat.
- **Insulin:** A key factor in determining insulin resistance and sensitivity.
- **BMI:** Heavily correlated with obesity-related diabetes onset.
- **Diabetes Pedigree Function:** Represents genetic risk; highly non-linear, ideal for fuzzy treatment.
- **Age:** Older patients have higher risk; risk progression can be mapped fuzzily across life stages.

These attributes will be fuzzified and mapped into membership functions to mimic how a doctor might say, “this patient is at high risk due to high glucose and elevated age.”

1.4 Sample List of Dataset

Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
6	148	72	35	0	33.6	0.627	50	1
1	85	66	29	0	26.6	0.351	31	0
8	183	64	0	0	23.3	0.672	32	1
1	89	66	23	94	28.1	0.167	21	0
0	137	40	35	168	43.1	2.288	33	1
5	116	74	0	0	25.6	0.201	30	0
3	78	50	32	88	31	0.248	26	1
10	115	0	0	0	35.3	0.134	29	0
2	197	70	45	543	30.5	0.158	53	1
8	125	96	0	0	0	0.232	54	1
4	110	92	0	0	37.6	0.191	30	0
10	168	74	0	0	38	0.537	34	1
10	139	80	0	0	27.1	1.441	57	0
1	189	60	23	846	30.1	0.398	59	1
5	166	72	19	175	25.8	0.587	51	1
7	100	0	0	0	30	0.484	32	1
0	118	84	47	230	45.8	0.551	31	1
7	107	74	0	0	29.6	0.254	31	1
1	103	30	38	83	43.3	0.183	33	0
1	115	70	30	96	34.6	0.529	32	1
3	126	88	41	235	39.3	0.704	27	0
8	99	84	0	0	35.4	0.388	50	0
7	196	90	0	0	39.8	0.451	41	1
9	119	80	35	0	29	0.263	29	1
11	143	94	33	146	36.6	0.254	51	1
10	125	70	26	115	31.1	0.205	41	1
7	147	76	0	0	39.4	0.257	43	1
1	97	66	15	140	23.2	0.487	22	0
13	145	82	19	110	22.2	0.245	57	0
5	117	92	0	0	34.1	0.337	38	0
5	109	75	26	0	36	0.546	60	0
3	158	76	36	245	31.6	0.851	28	1
3	88	58	11	54	24.8	0.267	22	0
6	92	92	0	0	19.9	0.188	28	0
10	122	78	31	0	27.6	0.512	45	0
4	103	60	33	192	24	0.966	33	0
11	138	76	0	0	33.2	0.42	35	0
9	102	76	37	0	32.9	0.665	46	1
2	90	68	42	0	38.2	0.503	27	1
4	111	72	47	207	37.1	1.39	56	1

3	180	64	25	70	34	0.271	26	0
7	133	84	0	0	40.2	0.696	37	0
7	106	92	18	0	22.7	0.235	48	0
9	171	110	24	240	45.4	0.721	54	1
7	159	64	0	0	27.4	0.294	40	0
0	180	66	39	0	42	1.893	25	1
1	146	56	0	0	29.7	0.564	29	0
2	71	70	27	0	28	0.586	22	0
7	103	66	32	0	39.1	0.344	31	1
7	105	0	0	0	0	0.305	24	0
1	103	80	11	82	19.4	0.491	22	0
1	101	50	15	36	24.2	0.526	26	0
5	88	66	21	23	24.4	0.342	30	0
8	176	90	34	300	33.7	0.467	58	1
7	150	66	42	342	34.7	0.718	42	0
1	73	50	10	0	23	0.248	21	0
7	187	68	39	304	37.7	0.254	41	1
0	100	88	60	110	46.8	0.962	31	0
0	146	82	0	0	40.5	1.781	44	0
0	105	64	41	142	41.5	0.173	22	0
2	84	0	0	0	0	0.304	21	0
8	133	72	0	0	32.9	0.27	39	1
5	44	62	0	0	25	0.587	36	0
2	141	58	34	128	25.4	0.699	24	0
7	114	66	0	0	32.8	0.258	42	1
5	99	74	27	0	29	0.203	32	0
0	109	88	30	0	32.5	0.855	38	1
2	109	92	0	0	42.7	0.845	54	0
1	95	66	13	38	19.6	0.334	25	0
4	146	85	27	100	28.9	0.189	27	0
2	100	66	20	90	32.9	0.867	28	1
5	139	64	35	140	28.6	0.411	26	0
13	126	90	0	0	43.4	0.583	42	1
4	129	86	20	270	35.1	0.231	23	0
1	79	75	30	0	32	0.396	22	0
1	0	48	20	0	24.7	0.14	22	0
7	62	78	0	0	32.6	0.391	41	0
5	95	72	33	0	37.7	0.37	27	0
0	131	0	0	0	43.2	0.27	26	1
2	112	66	22	0	25	0.307	24	0
3	113	44	13	0	22.4	0.14	22	0
2	74	0	0	0	0	0.102	22	0
7	83	78	26	71	29.3	0.767	36	0

0	101	65	28	0	24.6	0.237	22	0
5	137	108	0	0	48.8	0.227	37	1
2	110	74	29	125	32.4	0.698	27	0
13	106	72	54	0	36.6	0.178	45	0
2	100	68	25	71	38.5	0.324	26	0
15	136	70	32	110	37.1	0.153	43	1
1	107	68	19	0	26.5	0.165	24	0
1	80	55	0	0	19.1	0.258	21	0
4	123	80	15	176	32	0.443	34	0
7	81	78	40	48	46.7	0.261	42	0
4	134	72	0	0	23.8	0.277	60	1
2	142	82	18	64	24.7	0.761	21	0
6	144	72	27	228	33.9	0.255	40	0
2	92	62	28	0	31.6	0.13	24	0
1	71	48	18	76	20.4	0.323	22	0
6	93	50	30	64	28.7	0.356	23	0
1	122	90	51	220	49.7	0.325	31	1
1	163	72	0	0	39	1.222	33	1
1	151	60	0	0	26.1	0.179	22	0
0	125	96	0	0	22.5	0.262	21	0
1	81	72	18	40	26.6	0.283	24	0
2	85	65	0	0	39.6	0.93	27	0
1	126	56	29	152	28.7	0.801	21	0
1	96	122	0	0	22.4	0.207	27	0
4	144	58	28	140	29.5	0.287	37	0
3	83	58	31	18	34.3	0.336	25	0
0	95	85	25	36	37.4	0.247	24	1
3	171	72	33	135	33.3	0.199	24	1
8	155	62	26	495	34	0.543	46	1
1	89	76	34	37	31.2	0.192	23	0
4	76	62	0	0	34	0.391	25	0
7	160	54	32	175	30.5	0.588	39	1
4	146	92	0	0	31.2	0.539	61	1
5	124	74	0	0	34	0.22	38	1
5	78	48	0	0	33.7	0.654	25	0
4	97	60	23	0	28.2	0.443	22	0
4	99	76	15	51	23.2	0.223	21	0
0	162	76	56	100	53.2	0.759	25	1
6	111	64	39	0	34.2	0.26	24	0
2	107	74	30	100	33.6	0.404	23	0
5	132	80	0	0	26.8	0.186	69	0
0	113	76	0	0	33.3	0.278	23	1
1	88	30	42	99	55	0.496	26	1

3	120	70	30	135	42.9	0.452	30	0
1	118	58	36	94	33.3	0.261	23	0
1	117	88	24	145	34.5	0.403	40	1
0	105	84	0	0	27.9	0.741	62	1
4	173	70	14	168	29.7	0.361	33	1
9	122	56	0	0	33.3	1.114	33	1
3	170	64	37	225	34.5	0.356	30	1
8	84	74	31	0	38.3	0.457	39	0
2	96	68	13	49	21.1	0.647	26	0
2	125	60	20	140	33.8	0.088	31	0
0	100	70	26	50	30.8	0.597	21	0
0	93	60	25	92	28.7	0.532	22	0
0	129	80	0	0	31.2	0.703	29	0
5	105	72	29	325	36.9	0.159	28	0
3	128	78	0	0	21.1	0.268	55	0
5	106	82	30	0	39.5	0.286	38	0
2	108	52	26	63	32.5	0.318	22	0
10	108	66	0	0	32.4	0.272	42	1
4	154	62	31	284	32.8	0.237	23	0
0	102	75	23	0	0	0.572	21	0
9	57	80	37	0	32.8	0.096	41	0
2	106	64	35	119	30.5	1.4	34	0
5	147	78	0	0	33.7	0.218	65	0
2	90	70	17	0	27.3	0.085	22	0
1	136	74	50	204	37.4	0.399	24	0
4	114	65	0	0	21.9	0.432	37	0
9	156	86	28	155	34.3	1.189	42	1

1.5 Journal Paper

Title: Mediative Fuzzy Logic of Sugeno-Tsk Model For The Diagnosis of Diabetes

Authors: Sharma, M. K., Dhiman, N., & Mishra, V. N.

Published: 2021

Reference Link: <https://doi.org/10.1088/1742-6596/1724/1/012028>

Summary: Sharma, Dhiman, and Mishra (2021) developed a Sugeno-TSK fuzzy logic system enhanced with Mediative Fuzzy Logic (MFL) for diabetes diagnosis using the Pima Indians Diabetes Database. Their system addressed limitations in traditional fuzzy logic by handling contradictory and incomplete medical information through both membership and non-membership functions. The study utilized five key input variables: glucose levels, insulin, BMI, diabetes pedigree function, and age, with diabetes mellitus risk as the output variable. All variables were categorized into low, medium, and high levels using triangular membership functions. The system implemented 28 primary fuzzy rules following IF-THEN structures, such as "IF Glucose is low AND Insulin is high THEN DM is high." While their approach used a more complex Sugeno-TSK model with contradiction handling, the fundamental variable selection, triangular membership function design, and rule construction patterns provide valuable insights for developing simplified Mamdani-type systems for diabetes classification.

Title: Implementation of Fuzzy Mamdani Method in Diabetes

Author: Theenadayalan, C., Srinivasan, K., & Sudha, T.

Published: 2021

Reference: Turkish Online Journal of Qualitative Inquiry (TOJQI), 12(6), 719-724.

Summary: Theenadayalan, Srinivasan, and Sudha (2021) implemented a Mamdani fuzzy inference system specifically for diabetes diagnosis, focusing on fundamental fuzzy logic operations and rule structures. Their study utilized primary data from diabetic patients aged 15-80 years with corresponding HBA1C values, employing triangular fuzzy numbers for membership function representation. The research demonstrated two key approaches: single rule with single antecedent using crisp input values, and single rule with multiple antecedents combining both fuzzy and crisp inputs. The system applied both Max-Min and Max-Product composition methods for fuzzy inference, where a crisp input of age 29 years yielded a membership weight of 0.5, and multiple antecedent rules with weights $w_1 = 0.8$ and $w_2 = 0.3$ produced outputs of 0.3 (Max-Min) and 0.24 (Max-Product) respectively. While the study provides a simplified implementation focusing on age and HBA1C parameters, it offers valuable insights into basic Mamdani rule construction, triangular membership function design, and composition method comparisons that are directly applicable to developing straightforward fuzzy logic systems for diabetes classification.

Title: A Clinical Decision Support System based on fuzzy rules and classification algorithms for monitoring the physiological parameters of type-2 diabetic patients

Author: Colella, Y., De Lauri, C., Improta, G., Rossano, L., Vecchione, D., Spinosa, T., Giordano, V., Verdoliva, C., & Santini, S.

Published: 2021

Reference Link: <https://doi.org/10.3934/mbe.2021135>

Summary: Colella et al. (2021) developed a comprehensive Clinical Decision Support System that integrates a Mamdani-type Fuzzy Inference System with machine learning classifiers for type-2 diabetes patient monitoring and classification. Their approach utilizes five easily measurable physiological parameters such as systolic blood pressure, heart rate, temperature, oxygen saturation, and blood sugar as inputs to a fuzzy system that outputs a risk assessment scale from 0-14. The Mamdani FIS employs trapezoidal membership functions for inputs and triangular functions for outputs, with 1,800 fuzzy rules developed in collaboration with medical experts to ensure clinical accuracy. The system was validated on 1,523 real patient records from an Italian diabetes center, achieving 93.5% classification accuracy when combined with ensemble learning techniques. The fuzzy component successfully categorized patients into normal, low-risk, and high-risk groups, with outputs that consistently matched clinical diagnoses across four detailed case studies. This work demonstrates the practical effectiveness of Mamdani-type systems in diabetes care and provides a valuable framework for implementing fuzzy logic in medical decision support systems using commonly available physiological measurements.

Title: A Mamdani Fuzzy Inference System for Comprehensive Diabetes Risk Evaluation

Author: Tuly, T. A., Bijoy, M. H. I., Shafin, T. A., Arefin, M. S., Hasan, M. Z., & Uddin, M. S.

Published: 2025

Reference Link: <https://doi.org/10.1109/ECCE64574.2025.11013995>

Summary:

Tuly et al. developed a comprehensive diabetes risk evaluation system using a Mamdani Fuzzy Inference System (MFIS) to address the challenges of uncertainty in medical data and provide interpretable risk assessments. Their system incorporates nine input parameters including BMI, glucose levels, cholesterol measurements, and blood pressure, utilizing trapezoidal and triangular membership functions to classify each parameter into linguistic categories such as "Normal," "High," and "Very high." The researchers implemented 20 fuzzy rules that combine multiple health indicators using logical operators, with the system outputting diabetes risk predictions across five categories from "Low" to "Very-high" risk. Using the Mean of Maxima (MOM) defuzzification method, the system was tested on a dataset of 442 samples and demonstrated its ability to capture non-linear relationships between health parameters and diabetes risk. The study highlights the practical advantages of MFIS in healthcare applications, particularly its ability to incorporate expert medical knowledge through linguistic variables and provide transparent decision-making processes, making it suitable for clinical decision support systems where interpretability is crucial for medical professionals.

In summary

The collective contributions of these studies illustrate the versatility and effectiveness of fuzzy logic systems in the diagnosis and management of diabetes. By leveraging fuzzy inference systems, the authors address the inherent uncertainties and complexities of medical data, providing interpretable risk assessments that can support clinical decision-making. The research spans various methodologies, from the implementation of basic Mamdani systems to more complex Sugeno-TSK models, showcasing a progression in the sophistication of fuzzy logic applications in healthcare. The integration of expert knowledge and real patient data further enhances the reliability and applicability of these systems, making them valuable tools in the fight against diabetes.

We conclude that there is a need to develop efficient fuzzy-based systems in order to increase the accuracy of diabetes diagnosis and management, since such systems are capable of coping with uncertainties in medical data and producing interpretable risk assessments, which can aid clinical decisions.

Sharma et al. (2021) proved the utility of Mediative Fuzzy Logic applied to a Sugeno-TSK fuzzy logic model to solve the existent contradictions in medical information, allowing diabetes risk assessment with greater accuracy.

In like manner, Theenadayalan et al. (2021) demonstrated the significance of a Mamdani fuzzy inference system with simple rules and membership functions for detection of diabetes risk using patient data.

Colella et al. (2021) more recently developed this area by developing a Mamdani-type fuzzy model and machine learning classifiers, demonstrating high classification accuracy using readily measurable physiological parameters, and emphasizing the potential application of fuzzy logic in clinical scenarios.

Lastly, Tuly et al. (2025) introduced the discussion by proposing a holistic diabetes risk assessment framework, capturing interplay between health indicators and underlining the interpretative aspect of fuzzy logic systems essential for clinical decision making.

2. Methodology

2.1 Fuzzy Logic Framework

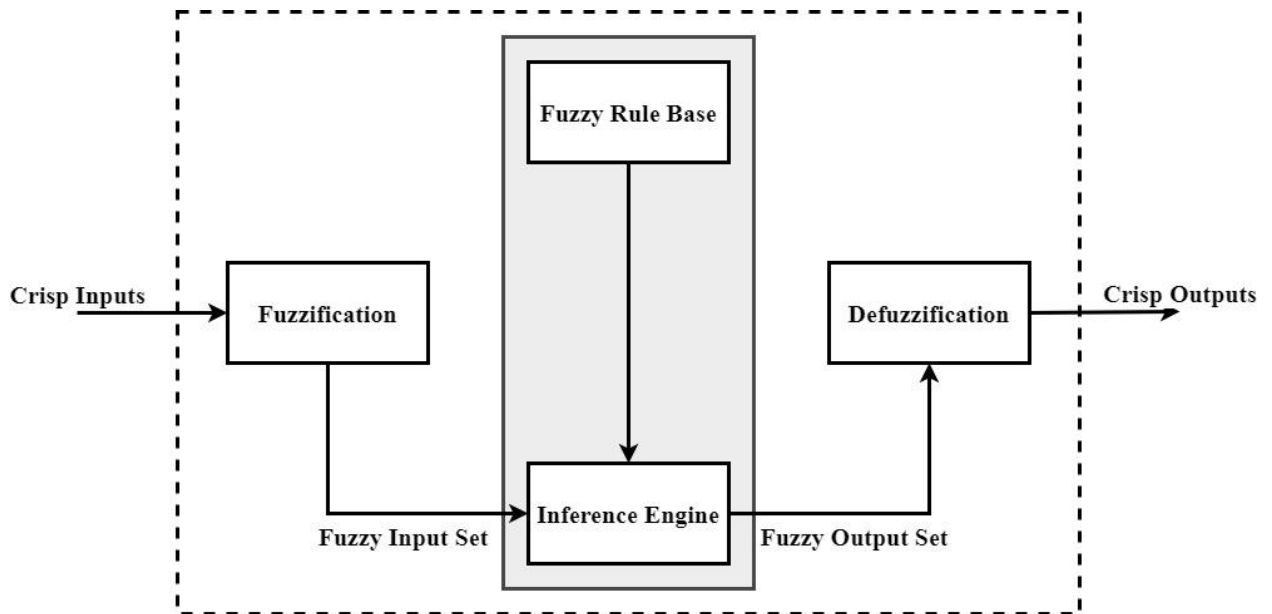


Figure 1: Fuzzy Logic Framework

Data Preprocessing:

The dataset was thoroughly analysed to see if any data creation was required before implementing the fuzzy logic framework into practice. AI systems frequently utilise preprocessing to clean data, deal with missing values, eliminate abnormalities, or normalise values for improved input handling. However, for the following reasons, it was decided that preprocessing was not required for this project:

1. No Missing or Invalid Values

Medical variables including blood pressure, BMI, insulin, and glucose may display zero values in several versions of the Pima Indians Diabetes dataset. These values are medically inaccurate (for example, a BMI of 0 is not possible). Nevertheless, in this particular dataset, no null or zero placeholders were identified in any of the eight chosen attributes and all input values were considered to be within legitimate, realistic ranges. Hence, no imputation was necessary, no outlier removal was needed, and data was clean enough to be used directly for fuzzification.

2. Fuzzy Systems Can Work with Raw Numerical Inputs

As long as the membership functions are constructed to fit the natural range of values, fuzzy logic systems can operate directly with real-world values (such as glucose in mg/dL or age in years), in contrast to machine learning models that frequently need normalised or scaled input.

3. Clarity and Integrity of System Design

All input variables have been preserved in their original units and fuzzified directly without change in order to maintain the system's focus and interpretability and to prevent bias or distortion through unnecessary preprocessing.

Stage 1: Crisp Inputs

The system draws in the dataset's **raw medical data** at this initial stage. For a single patient, they are **numerical (crisp) values** that indicate certain health metrics. Here, the system only receives data where there is no processing, transformation, or logic takes place.

Crisp Definition

In fuzzy logic, "crisp" refers to precise, clear values that are neither fuzzy nor unclear. These actual figures come from the dataset itself.

Importance of Crisp Inputs

The system's basic numerical base is provided by this step. These clear values provide the foundation for all subsequent processes, including fuzzification, reasoning, and decision-making.

Stage 2: Fuzzification

Each **Membership functions** are utilised to **convert** every separate input value **into fuzzy values**. They show how much a value falls into one or more **linguistic categories**.

For example:

- Glucose = 135 → might be 0.6 *Medium* and 0.4 *High*
- Age = 28 → might be 0.9 *Young*, 0.1 *Middle-aged*

Membership Functions Definition

Membership functions are **mathematical shapes** which is often shaped as trapezoids or triangles, that are used to provide input values membership levels ranging from 0 to 1.

Each input has **3 fuzzy sets**:

- Example for BMI : *Low, Medium, High*
- Example for Age : *Young, Middle-aged, Old*

Importance of Fuzzification

Fuzzification allows the system to reason like a human:

“This patient is not just ‘BMI = 31’, they are *mostly overweight*, but *partly normal*.”

Just as doctors evaluate health data in ranges, it converts fixed numbers into **relevant fuzzy categories**.

Stage 3: Fuzzy Rule Base

All of the **medical logic** is kept here in the form of **IF-THEN rules**. Logical assumptions founded on expert knowledge or journal-based patterns are included in the fuzzy rule base.

Each rule connects input conditions to a fuzzy output.

Example Rules:

- Rule 1: IF Glucose is *High* AND BMI is *High* THEN Risk is *High*
- Rule 2: IF Age is *Young* AND Glucose is *Low* THEN Risk is *Low*
- Rule 3: IF Insulin is *Low* AND Blood Pressure is *Normal* THEN Risk is *Medium*

Structure of Rules

Each rule uses fuzzy inputs and applies logical connectors:

- **AND** = minimum of values
- **OR** = maximum of values

All legitimate outputs are retained for aggregate, and the system can **employ numerous rules simultaneously**.

Importance of Fuzzy Rule Base

This step contains the "**expert knowledge**" that gives your fuzzy system its intelligent behaviour. The reasoning contained here is a reflection of **how actual doctors** use a combination of risk factors and symptoms to **make choices**.

Stage 4: Inference Engine

All active rules are processed by the inference engine, which then uses the fuzzy inputs to determine how strongly each rule applies before combining the results into a single **fuzzy output set**.

The Process

Let's say:

- Rule A set off with strength 0.6 → suggests *High Risk*
- Rule B set off with strength 0.4 → suggests *Medium Risk*

These outputs are **blended** to form a fuzzy output shape. This is where the system applies:

- **MIN operator** for AND conditions,
- **MAX operator** to aggregate overlapping outputs from multiple rules.

Importance of Inference Engine

All the reasoning takes place there. The inference engine simulates clinical decision making, in which a variety of factors are taken into consideration before reaching a final conclusion. Instead of a single label, it produces a fuzzy judgement, such as:

“The patient is 60% high risk, 40% medium risk.”

Stage 5: Fuzzy Output Set

Following the processing of each rule, the system generates a fuzzy output shape that indicates the collective decision made by all of the rules. This is not yet a number but it is a verbal conclusion.

Example:

The fuzzy output might be a mix like:

- 30% *Low Risk*
- 50% *Medium Risk*
- 70% *High Risk*

This overlap reflects **real-world uncertainty** in medical diagnosis.

Importance of Fuzzy Output Set

The system does not make quick judgements. Like a medical professional, it provides a risk range rather than a binary response, allowing for **partial truths**.

Stage 6: Defuzzification → Crisp Output

The **Centroid approach** (the shape's centre of gravity) **transforms the fuzzy output set** (such as the overlapping hazards) **into a single, distinct value**. This provides with a **final risk score**, usually in the range of 0 and 1.

Example:

- Fuzzy output shape → Centroid = 0.67
- Interpretation:
 - $0.67 \geq 0.5$ → **Classified as Diabetic (1)**
 - Value < 0.5 → **Classified as Non-Diabetic (0)**

This result is then added to a new column called Fuzzy_Result.

Importance of Defuzzification

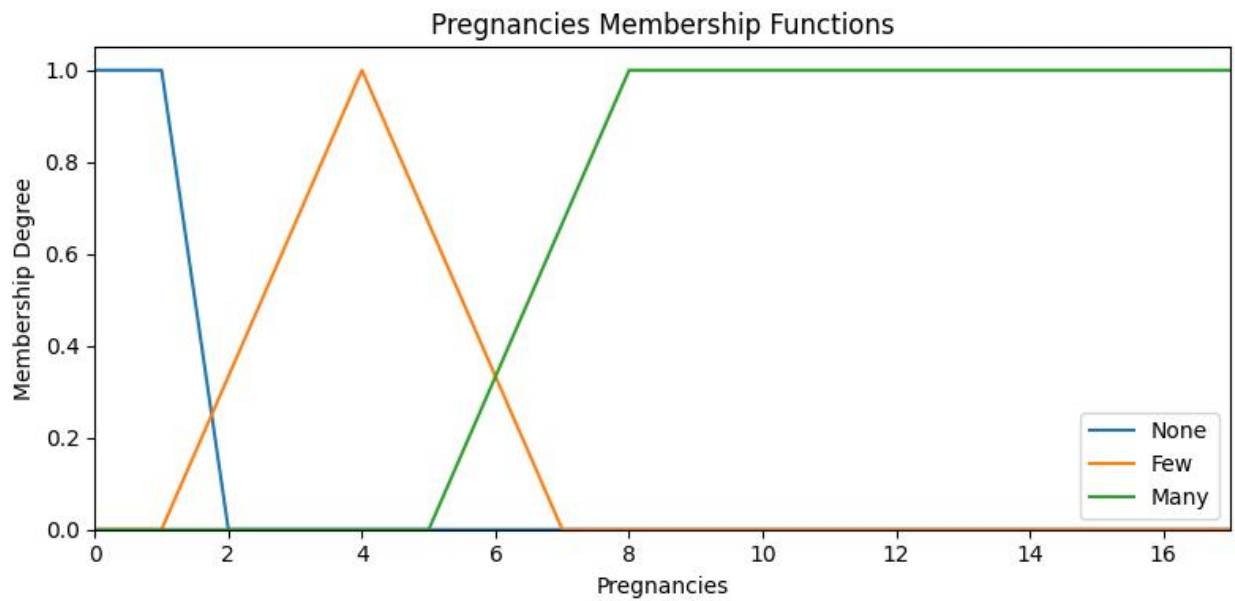
Fuzzy systems use vague reasoning, while real-world judgements require **precise answers**. A vague output can be defuzzied to produce a **precise result** that can be used for performance evaluation and classification.

2.2 Fuzzy Logic Membership Functions (Input And Output)

Input Variable 1: Pregnancies

Increased diabetes risk is connected to more pregnancies (due to hormonal changes and gestational diabetes).

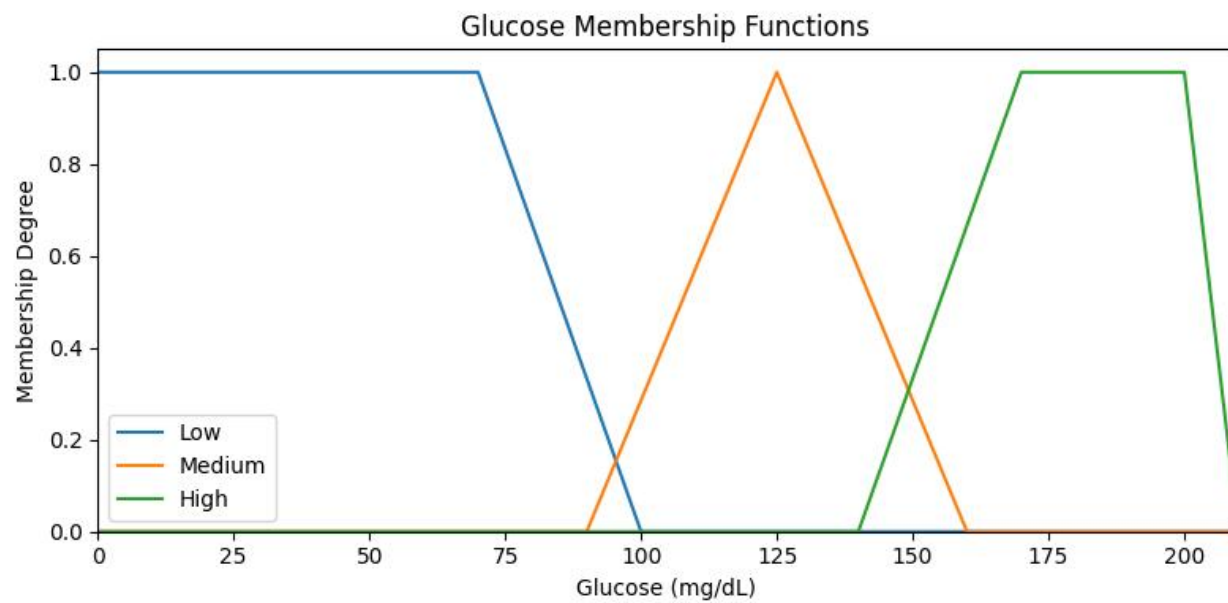
```
pregnancies['None'] = fuzz.trapmf(pregnancies.universe, [0, 0, 1, 2])  
pregnancies['Few'] = fuzz.trimf(pregnancies.universe, [1, 4, 7])  
pregnancies['Many'] = fuzz.trapmf(pregnancies.universe, [5, 8, 17, 17])
```



Input Variable 2: Glucose

The most obvious sign of diabetes is blood sugar levels.

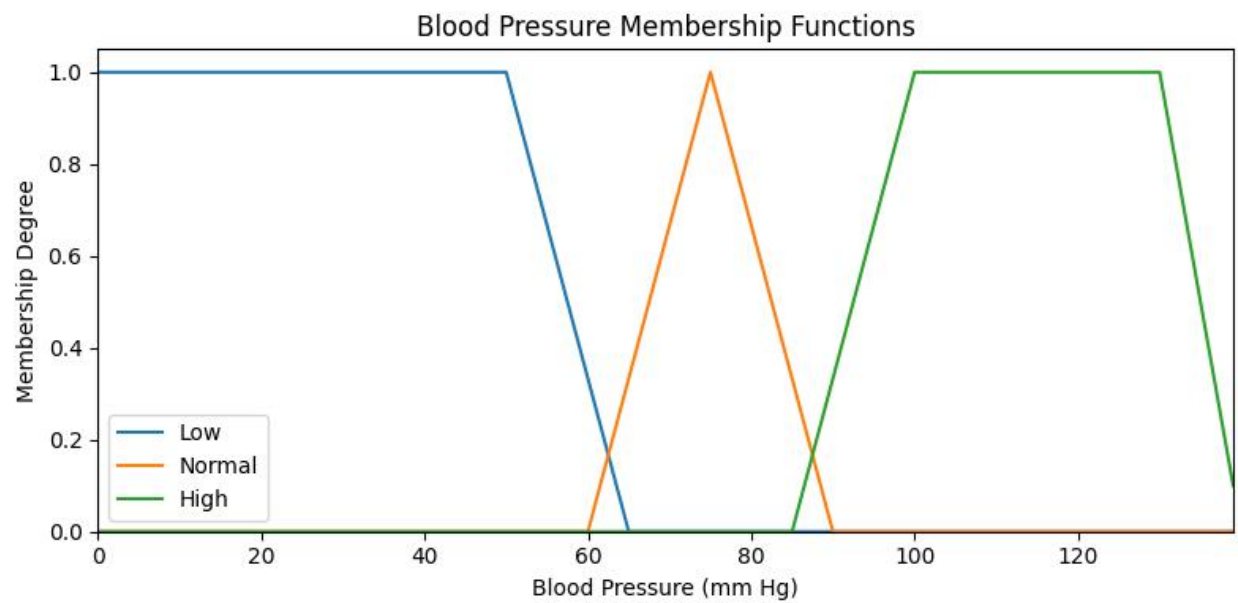
```
glucose['Low'] = fuzz.trapmf(glucose.universe, [0, 0, 70, 100])  
glucose['Medium'] = fuzz.trimf(glucose.universe, [90, 125, 160])  
glucose['High'] = fuzz.trapmf(glucose.universe, [140, 170, 200, 210])
```



Input Variable 3: Blood Pressure

Diabetes frequently co-occurs with high blood pressure.

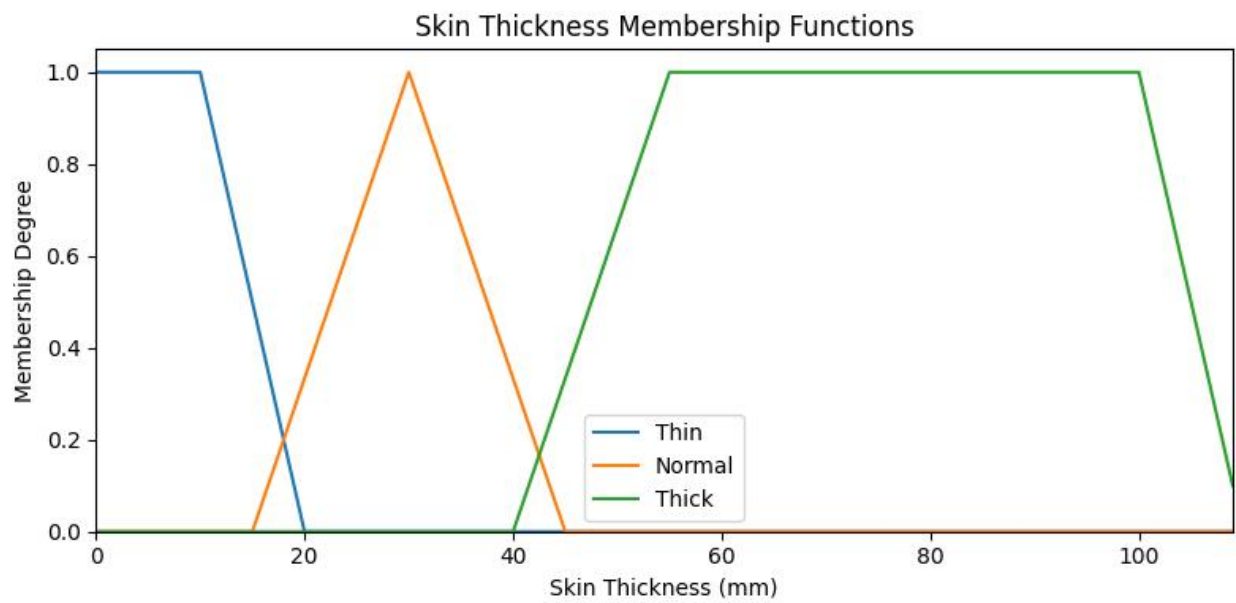
```
blood_pressure['Low'] = fuzz.trapmf(blood_pressure.universe, [0, 0, 50, 65])  
blood_pressure['Normal'] = fuzz.trimf(blood_pressure.universe, [60, 75, 90])  
blood_pressure['High'] = fuzz.trapmf(blood_pressure.universe, [85, 100, 130, 140])
```



Input Variable 4: Skin Thickness

Shows the amount of body fat, which is correlated with insulin resistance.

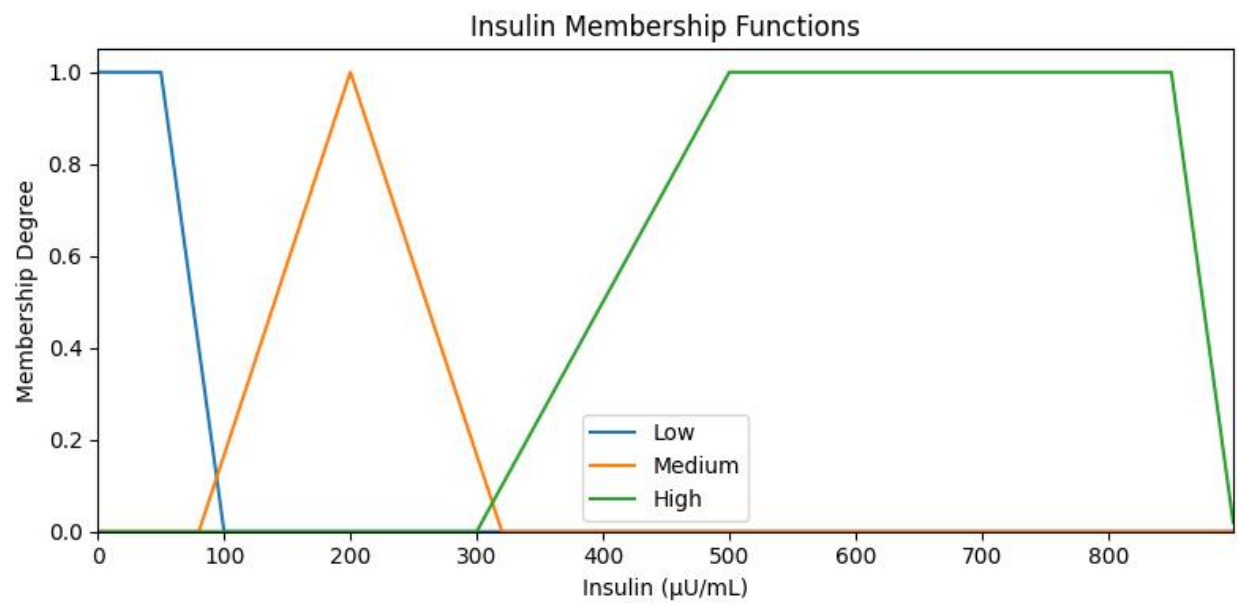
```
skin_thickness['Thin'] = fuzz.trapmf(skin_thickness.universe, [0, 0, 10, 20])  
skin_thickness['Normal'] = fuzz.trimf(skin_thickness.universe, [15, 30, 45])  
skin_thickness['Thick'] = fuzz.trapmf(skin_thickness.universe, [40, 55, 100, 110])
```



Input Variable 5: Insulin

High insulin levels are evidence of insulin resistance, a key feature of diabetes.

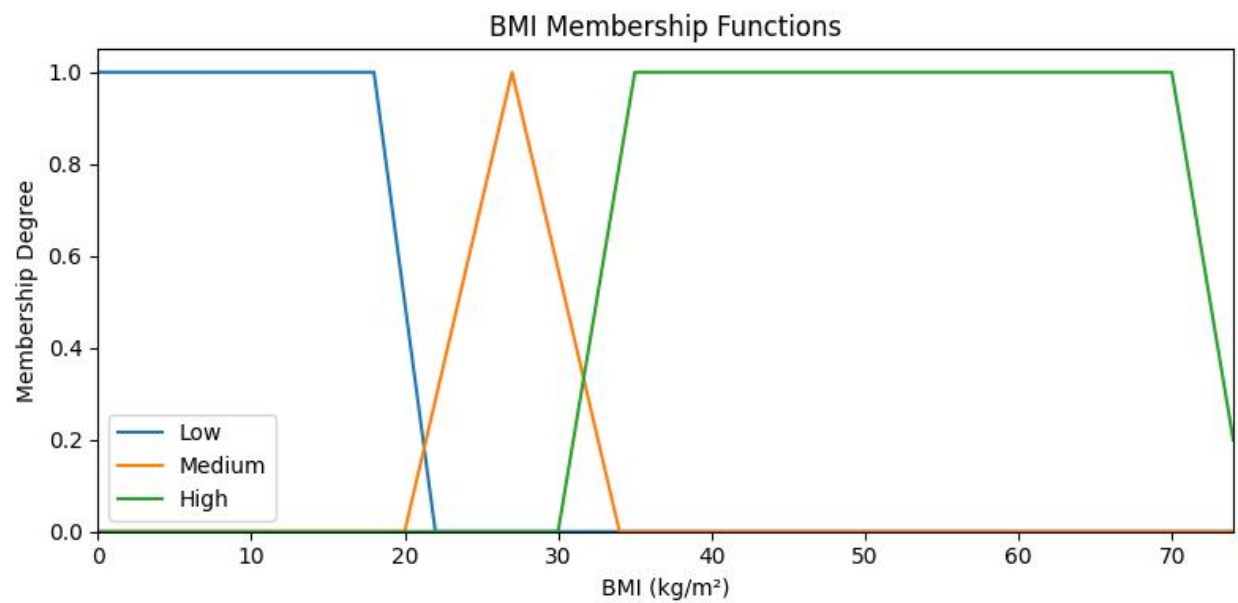
```
insulin['Low'] = fuzz.trapmf(insulin.universe, [0, 0, 50, 100])  
insulin['Medium'] = fuzz.trimf(insulin.universe, [80, 200, 320])  
insulin['High'] = fuzz.trapmf(insulin.universe, [300, 500, 850, 900])
```



Input Variable 6: BMI

One well-known risk factor for Type 2 diabetes is having a high body mass index.

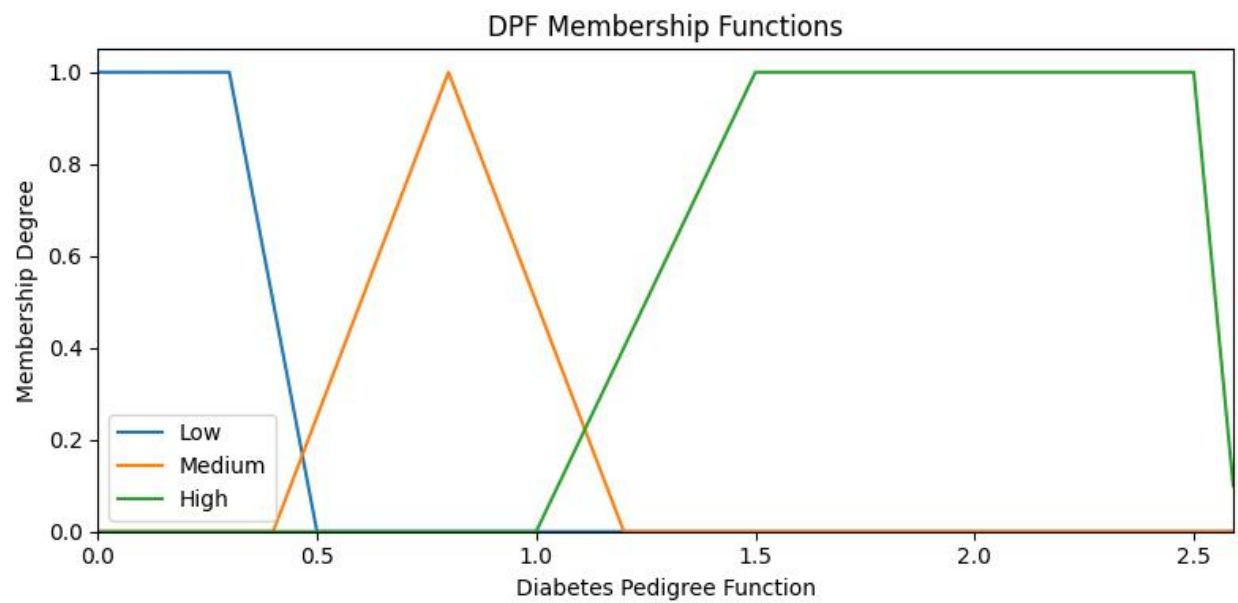
```
bmi['Low'] = fuzz.trapmf(bmi.universe, [0, 0, 18, 22])  
bmi['Medium'] = fuzz.trimf(bmi.universe, [20, 27, 34])  
bmi['High'] = fuzz.trapmf(bmi.universe, [30, 35, 70, 75])
```



Input Variable 7: Diabetes Pedigree Function (DPF)

Stronger family histories of diabetes are indicated by higher readings.

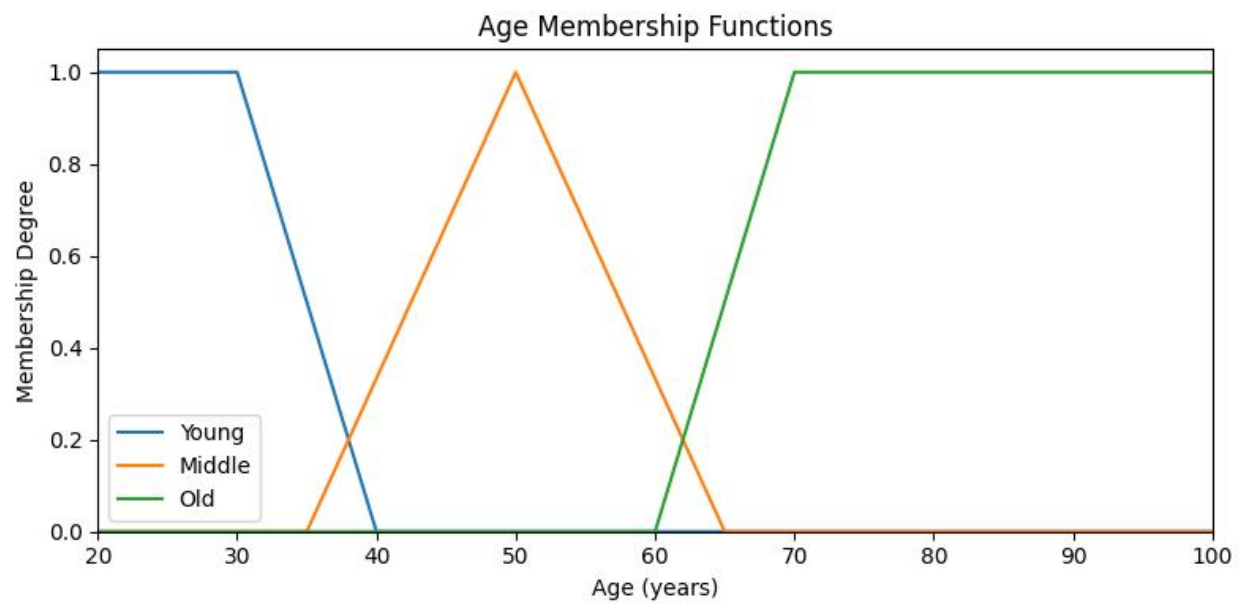
```
dpf['Low'] = fuzz.trapmf(dpf.universe, [0.0, 0.0, 0.3, 0.5])  
dpf['Medium'] = fuzz.trimf(dpf.universe, [0.4, 0.8, 1.2])  
dpf['High'] = fuzz.trapmf(dpf.universe, [1.0, 1.5, 2.5, 2.6])
```



Input Variable 8: Age

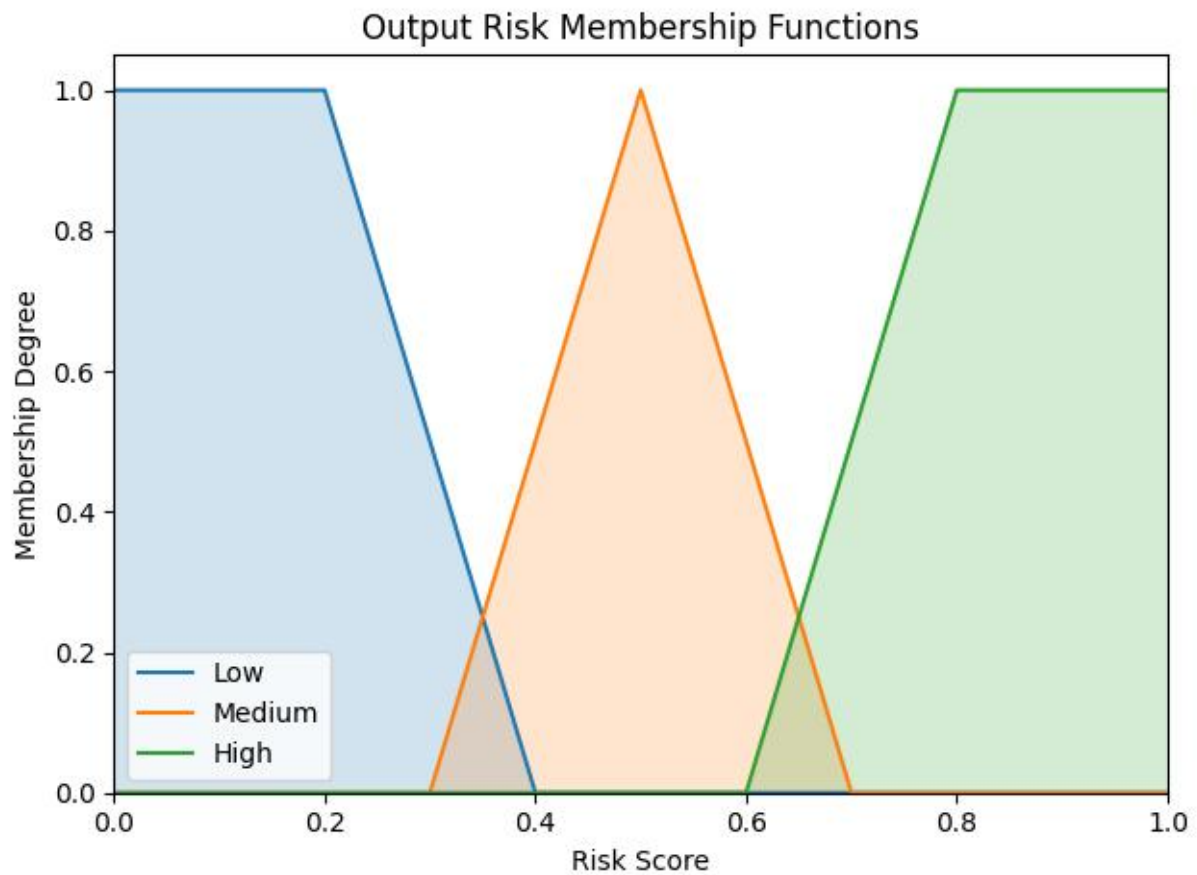
As people age, their metabolism slows down, increasing their risk of diabetes.

```
age['Young'] = fuzz.trapmf(age.universe, [20, 20, 30, 40])  
age['Middle'] = fuzz.trimf(age.universe, [35, 50, 65])  
age['Old'] = fuzz.trapmf(age.universe, [60, 70, 100, 100])
```



Output Variable: Diabetes Risk

```
risk['Low'] = fuzz.trapmf(risk.universe, [0.0, 0.0, 0.2, 0.4])  
risk['Medium'] = fuzz.trimf(risk.universe, [0.3, 0.5, 0.7])  
risk['High'] = fuzz.trapmf(risk.universe, [0.6, 0.8, 1.0, 1.0])
```



2.3 Fuzzy Logic Rules

High Risk Rules (Strong Diabetes Indicators)

Rule No.	Rule
1	IF Glucose is High AND BMI is High THEN Risk is High
2	IF Glucose is High AND Age is Old THEN Risk is High
3	IF Glucose is High AND Insulin is High THEN Risk is High
4	IF Glucose is High AND DPF is High THEN Risk is High
5	IF Pregnancies are Many AND Glucose is High THEN Risk is High
6	IF BMI is High AND DPF is High THEN Risk is High
7	IF Glucose is High AND Age is Middle-aged AND BMI is High THEN Risk is High
8	IF BloodPressure is High AND BMI is High THEN Risk is High
9	IF Insulin is High AND Age is Old THEN Risk is High
10	IF Glucose is High AND SkinThickness is Thick AND Insulin is High THEN Risk is High
11	IF Glucose is High AND BMI is High AND Pregnancies are Many THEN Risk is High
12	IF DPF is High AND Age is Old THEN Risk is High
13	IF BMI is High AND SkinThickness is Thick THEN Risk is High

Medium Risk Rules (Borderline / Contributing Factors Present)

Rule No.	Rule
14	IF Glucose is Medium AND BMI is Medium THEN Risk is Medium
15	IF Glucose is Medium AND Age is Middle-aged THEN Risk is Medium
16	IF BMI is Medium AND DPF is Medium THEN Risk is Medium
17	IF Pregnancies are Few AND Glucose is Medium THEN Risk is Medium
18	IF Insulin is Medium AND Glucose is Medium THEN Risk is Medium
19	IF BloodPressure is Normal AND BMI is Medium THEN Risk is Medium
20	IF Age is Middle-aged AND DPF is Medium THEN Risk is Medium
21	IF SkinThickness is Normal AND BMI is Medium THEN Risk is Medium
22	IF BloodPressure is High AND Glucose is Medium THEN Risk is Medium
23	IF Pregnancies are Many AND Age is Middle-aged THEN Risk is Medium
24	IF Glucose is Medium AND DPF is High THEN Risk is Medium
25	IF Insulin is Medium AND BMI is Medium THEN Risk is Medium
26	IF Glucose is High AND DPF is Medium THEN Risk is Medium
27	IF SkinThickness is Thick AND BMI is Medium THEN Risk is Medium

Low Risk Rules (Healthy Patterns)

Rule No.	Rule
28	IF Glucose is Low AND BMI is Low THEN Risk is Low
29	IF Glucose is Low AND Age is Young THEN Risk is Low
30	IF Glucose is Low AND DPF is Low THEN Risk is Low
31	IF Pregnancies are None AND Glucose is Low THEN Risk is Low
32	IF BloodPressure is Normal AND Glucose is Low THEN Risk is Low
33	IF BMI is Low AND SkinThickness is Thin THEN Risk is Low
34	IF Insulin is Low AND Glucose is Low THEN Risk is Low
35	IF DPF is Low AND Age is Young THEN Risk is Low
36	IF Insulin is Low AND BMI is Low THEN Risk is Low
37	IF Pregnancies are None AND DPF is Low AND Glucose is Low THEN Risk is Low
38	IF Age is Young AND BMI is Low THEN Risk is Low
39	IF Glucose is Low AND SkinThickness is Thin THEN Risk is Low
40	IF BloodPressure is Low AND Glucose is Low THEN Risk is Low

3. Results

3.1 Classification Results

Pregnancies	Glucose	BloodPressure	Skin Thickness	Insulin	BMI	Diabetes Pedigree Function	Age	Outcome	Risk_Score	Fuzzy_Result	Fuzzy_Classification
6	148	72	35	0	33.6	0.627	50	1	0.617317	1	Diabetic
1	85	66	29	0	26.6	0.351	31	0	0.315389	0	Non-Diabetic
8	183	64	0	0	23.3	0.672	32	1	0.717246	1	Diabetic
1	89	66	23	94	28.1	0.167	21	0	0.271462	0	Non-Diabetic
0	137	40	35	168	43.1	2.288	33	1	0.718665	1	Diabetic
5	116	74	0	0	25.6	0.201	30	0	0.288408	0	Non-Diabetic
3	78	50	32	88	31	0.248	26	1	0.259688	0	Non-Diabetic
10	115	0	0	0	35.3	0.134	29	0	0.155556	0	Non-Diabetic
2	197	70	45	543	30.5	0.158	53	1	0.731905	1	Diabetic
8	125	96	0	0	0	0.232	54	1	0.285626	0	Non-Diabetic
4	110	92	0	0	37.6	0.191	30	0	0.41679	0	Non-Diabetic
10	168	74	0	0	38	0.537	34	1	0.747878	1	Diabetic
10	139	80	0	0	27.1	1.441	57	0	0.5	0	Non-Diabetic
1	189	60	23	846	30.1	0.398	59	1	0.728538	1	Diabetic
5	166	72	19	175	25.8	0.587	51	1	0.5	0	Non-Diabetic
7	100	0	0	0	30	0.484	32	1	0.430412	0	Non-Diabetic
0	118	84	47	230	45.8	0.551	31	1	0.650843	1	Diabetic
7	107	74	0	0	29.6	0.254	31	1	0.287735	0	Non-Diabetic
1	103	30	38	83	43.3	0.183	33	0	0.179482	0	Non-Diabetic
1	115	70	30	96	34.6	0.529	32	1	0.5	1	Diabetic
3	126	88	41	235	39.3	0.704	27	0	0.588048	1	Diabetic
8	99	84	0	0	35.4	0.388	50	0	0.483191	0	Non-Diabetic
7	196	90	0	0	39.8	0.451	41	1	0.744184	1	Diabetic
9	119	80	35	0	29	0.263	29	1	0.284669	0	Non-Diabetic
11	143	94	33	146	36.6	0.254	51	1	0.666624	1	Diabetic
10	125	70	26	115	31.1	0.205	41	1	0.5	1	Diabetic

7	147	76	0	0	39.4	0.257	43	1	0.609995	1	Diabetic
1	97	66	15	140	23.2	0.487	22	0	0.437852	0	Non-Diabetic
13	145	82	19	110	22.2	0.245	57	0	0.586172	1	Diabetic
5	117	92	0	0	34.1	0.337	38	0	0.5749	1	Diabetic
5	109	75	26	0	36	0.546	60	0	0.5	0	Non-Diabetic
3	158	76	36	245	31.6	0.851	28	1	0.629117	1	Diabetic
3	88	58	11	54	24.8	0.267	22	0	0.155556	0	Non-Diabetic
6	92	92	0	0	19.9	0.188	28	0	0.176588	0	Non-Diabetic
10	122	78	31	0	27.6	0.512	45	0	0.5	0	Non-Diabetic
4	103	60	33	192	24	0.966	33	0	0.5	1	Diabetic
11	138	76	0	0	33.2	0.42	35	0	0.24959	0	Non-Diabetic
9	102	76	37	0	32.9	0.665	46	1	0.5	0	Non-Diabetic
2	90	68	42	0	38.2	0.503	27	1	0.368593	0	Non-Diabetic
4	111	72	47	207	37.1	1.39	56	1	0.702816	1	Diabetic
3	180	64	25	70	34	0.271	26	0	0.469161	0	Non-Diabetic
7	133	84	0	0	40.2	0.696	37	0	0.5	1	Diabetic
7	106	92	18	0	22.7	0.235	48	0	0.5	0	Non-Diabetic
9	171	110	24	240	45.4	0.721	54	1	0.711504	1	Diabetic
7	159	64	0	0	27.4	0.294	40	0	0.719413	1	Diabetic
0	180	66	39	0	42	1.893	25	1	0.844444	1	Diabetic
1	146	56	0	0	29.7	0.564	29	0	0.5	0	Non-Diabetic
2	71	70	27	0	28	0.586	22	0	0.291085	0	Non-Diabetic
7	103	66	32	0	39.1	0.344	31	1	0.164149	0	Non-Diabetic
7	105	0	0	0	0	0.305	24	0	0.155556	0	Non-Diabetic
1	103	80	11	82	19.4	0.491	22	0	0.178157	0	Non-Diabetic
1	101	50	15	36	24.2	0.526	26	0	0.5	1	Diabetic
5	88	66	21	23	24.4	0.342	30	0	0.274047	0	Non-Diabetic
8	176	90	34	300	33.7	0.467	58	1	0.735615	1	Diabetic
7	150	66	42	342	34.7	0.718	42	0	0.64542	1	Diabetic
1	73	50	10	0	23	0.248	21	0	0.155556	0	Non-Diabetic
7	187	68	39	304	37.7	0.254	41	1	0.744184	1	Diabetic

0	100	88	60	110	46.8	0.962	31	0	0.770556	1	Diabetic
0	146	82	0	0	40.5	1.781	44	0	0.744184	1	Diabetic
0	105	64	41	142	41.5	0.173	22	0	0.286848	0	Non-Diabetic
2	84	0	0	0	0	0.304	21	0	0.155556	0	Non-Diabetic
8	133	72	0	0	32.9	0.27	39	1	0.413375	0	Non-Diabetic
5	44	62	0	0	25	0.587	36	0	0.264482	0	Non-Diabetic
2	141	58	34	128	25.4	0.699	24	0	0.5	0	Non-Diabetic
7	114	66	0	0	32.8	0.258	42	1	0.5	0	Non-Diabetic
5	99	74	27	0	29	0.203	32	0	0.302629	0	Non-Diabetic
0	109	88	30	0	32.5	0.855	38	1	0.654747	1	Diabetic
2	109	92	0	0	42.7	0.845	54	0	0.651611	1	Diabetic
1	95	66	13	38	19.6	0.334	25	0	0.162122	0	Non-Diabetic
4	146	85	27	100	28.9	0.189	27	0	0.285394	0	Non-Diabetic
2	100	66	20	90	32.9	0.867	28	1	0.5	0	Non-Diabetic
5	139	64	35	140	28.6	0.411	26	0	0.348652	0	Non-Diabetic
13	126	90	0	0	43.4	0.583	42	1	0.64542	1	Diabetic
4	129	86	20	270	35.1	0.231	23	0	0.31329	0	Non-Diabetic
1	79	75	30	0	32	0.396	22	0	0.263197	0	Non-Diabetic
1	0	48	20	0	24.7	0.14	22	0	0.24547	0	Non-Diabetic
7	62	78	0	0	32.6	0.391	41	0	0.255816	0	Non-Diabetic
5	95	72	33	0	37.7	0.37	27	0	0.229826	0	Non-Diabetic
0	131	0	0	0	43.2	0.27	26	1	0.155556	0	Non-Diabetic
2	112	66	22	0	25	0.307	24	0	0.282189	0	Non-Diabetic
3	113	44	13	0	22.4	0.14	22	0	0.281335	0	Non-Diabetic
2	74	0	0	0	0	0.102	22	0	0.155556	0	Non-Diabetic
7	83	78	26	71	29.3	0.767	36	0	0.328829	0	Non-Diabetic
0	101	65	28	0	24.6	0.237	22	0	0.281335	0	Non-Diabetic
5	137	108	0	0	48.8	0.227	37	1	0.621626	1	Diabetic
2	110	74	29	125	32.4	0.698	27	0	0.5	0	Non-Diabetic
13	106	72	54	0	36.6	0.178	45	0	0.712611	1	Diabetic
2	100	68	25	71	38.5	0.324	26	0	0.246019	0	Non-Diabetic

15	136	70	32	110	37.1	0.153	43	1	0.5	0	Non-Diabetic
1	107	68	19	0	26.5	0.165	24	0	0.271462	0	Non-Diabetic
1	80	55	0	0	19.1	0.258	21	0	0.155556	0	Non-Diabetic
4	123	80	15	176	32	0.443	34	0	0.392874	0	Non-Diabetic
7	81	78	40	48	46.7	0.261	42	0	0.301007	0	Non-Diabetic
4	134	72	0	0	23.8	0.277	60	1	0.5	0	Non-Diabetic
2	142	82	18	64	24.7	0.761	21	0	0.5	1	Diabetic
6	144	72	27	228	33.9	0.255	40	0	0.578457	1	Diabetic
2	92	62	28	0	31.6	0.13	24	0	0.24707	0	Non-Diabetic
1	71	48	18	76	20.4	0.323	22	0	0.178224	0	Non-Diabetic
6	93	50	30	64	28.7	0.356	23	0	0.313403	0	Non-Diabetic
1	122	90	51	220	49.7	0.325	31	1	0.48112	0	Non-Diabetic
1	163	72	0	0	39	1.222	33	1	0.835304	1	Diabetic
1	151	60	0	0	26.1	0.179	22	0	0.230974	0	Non-Diabetic
0	125	96	0	0	22.5	0.262	21	0	0.285626	0	Non-Diabetic
1	81	72	18	40	26.6	0.283	24	0	0.288408	0	Non-Diabetic
2	85	65	0	0	39.6	0.93	27	0	0.17619	0	Non-Diabetic
1	126	56	29	152	28.7	0.801	21	0	0.5	0	Non-Diabetic
1	96	122	0	0	22.4	0.207	27	0	0.210797	0	Non-Diabetic
4	144	58	28	140	29.5	0.287	37	0	0.378952	0	Non-Diabetic
3	83	58	31	18	34.3	0.336	25	0	0.162524	0	Non-Diabetic
0	95	85	25	36	37.4	0.247	24	1	0.155556	0	Non-Diabetic
3	171	72	33	135	33.3	0.199	24	1	0.443946	0	Non-Diabetic
8	155	62	26	495	34	0.543	46	1	0.657322	1	Diabetic
1	89	76	34	37	31.2	0.192	23	0	0.255816	0	Non-Diabetic
4	76	62	0	0	34	0.391	25	0	0.163333	0	Non-Diabetic
7	160	54	32	175	30.5	0.588	39	1	0.699335	1	Diabetic
4	146	92	0	0	31.2	0.539	61	1	0.627629	1	Diabetic
5	124	74	0	0	34	0.22	38	1	0.410055	0	Non-Diabetic
5	78	48	0	0	33.7	0.654	25	0	0.18544	0	Non-Diabetic
4	97	60	23	0	28.2	0.443	22	0	0.374477	0	Non-Diabetic

4	99	76	15	51	23.2	0.223	21	0	0.263258	0	Non-Diabetic
0	162	76	56	100	53.2	0.759	25	1	0.714374	1	Diabetic
6	111	64	39	0	34.2	0.26	24	0	0.24547	0	Non-Diabetic
2	107	74	30	100	33.6	0.404	23	0	0.303104	0	Non-Diabetic
5	132	80	0	0	26.8	0.186	69	0	0.5	0	Non-Diabetic
0	113	76	0	0	33.3	0.278	23	1	0.190505	0	Non-Diabetic
1	88	30	42	99	55	0.496	26	1	0.345737	0	Non-Diabetic
3	120	70	30	135	42.9	0.452	30	0	0.397229	0	Non-Diabetic
1	118	58	36	94	33.3	0.261	23	0	0.195577	0	Non-Diabetic
1	117	88	24	145	34.5	0.403	40	1	0.597964	1	Diabetic
0	105	84	0	0	27.9	0.741	62	1	0.5	0	Non-Diabetic
4	173	70	14	168	29.7	0.361	33	1	0.307825	0	Non-Diabetic
9	122	56	0	0	33.3	1.114	33	1	0.661272	1	Diabetic
3	170	64	37	225	34.5	0.356	30	1	0.531665	1	Diabetic
8	84	74	31	0	38.3	0.457	39	0	0.279978	0	Non-Diabetic
2	96	68	13	49	21.1	0.647	26	0	0.318685	0	Non-Diabetic
2	125	60	20	140	33.8	0.088	31	0	0.276315	0	Non-Diabetic
0	100	70	26	50	30.8	0.597	21	0	0.5	0	Non-Diabetic
0	93	60	25	92	28.7	0.532	22	0	0.39927	0	Non-Diabetic
0	129	80	0	0	31.2	0.703	29	0	0.5	0	Non-Diabetic
5	105	72	29	325	36.9	0.159	28	0	0.259688	0	Non-Diabetic
3	128	78	0	0	21.1	0.268	55	0	0.401873	0	Non-Diabetic
5	106	82	30	0	39.5	0.286	38	0	0.39395	0	Non-Diabetic
2	108	52	26	63	32.5	0.318	22	0	0.252449	0	Non-Diabetic
10	108	66	0	0	32.4	0.272	42	1	0.5	0	Non-Diabetic
4	154	62	31	284	32.8	0.237	23	0	0.400651	0	Non-Diabetic
0	102	75	23	0	0	0.572	21	0	0.155556	0	Non-Diabetic
9	57	80	37	0	32.8	0.096	41	0	0.255816	0	Non-Diabetic
2	106	64	35	119	30.5	1.4	34	0	0.559079	1	Diabetic
5	147	78	0	0	33.7	0.218	65	0	0.62985	1	Diabetic
2	90	70	17	0	27.3	0.085	22	0	0.28194	0	Non-Diabetic

1	136	74	50	204	37.4	0.399	24	0	0.529585	1	Diabetic
4	114	65	0	0	21.9	0.432	37	0	0.381451	0	Non-Diabetic
9	156	86	28	155	34.3	1.189	42	1	0.684688	1	Diabetic

3.2 Classification Accuracy

The anticipated results (Fuzzy_Result) and the actual medical outcomes (Outcome) from the dataset were compared in order to assess the fuzzy logic system's performance. To assess the system output, the usual accuracy formula was used:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$

Evaluate Classification Accuracy

```
✓ [86] total_predictions = len(df_cleaned)
      correct_predictions = (df_cleaned['Fuzzy_Result'] == df_cleaned['Outcome']).sum()
      accuracy = (correct_predictions / total_predictions) * 100
```

Output Summary

```
✓ [87] print(f"Total Predictions: {total_predictions}")
      print(f"Correct Predictions: {correct_predictions}")
      print(f"Accuracy: {accuracy:.2f}%")
```

```
➡ Total Predictions: 768
   Correct Predictions: 558
   Accuracy: 72.66%
```

Classification Accuracy

This outcome shows that the fuzzy logic system can accurately categorise diabetes risk while preserving complete explainability. The **72.66%** accuracy attained shows good performance for a rule-based AI system that was manually created without training or optimisation.

4. Conclusion

This project successfully implemented a **fuzzy logic-based classification system** to assess diabetes risk using the **Kaggle Pima Indians Diabetes Dataset**. The system utilised eight clinical input variables, including glucose, insulin, age, BMI, and others, which were mapped to linguistic fuzzy categories such as Low, Medium, and High. The output variable, Risk, was similarly modelled with fuzzy sets and converted into a crisp value through **Centroid Defuzzification**.

To simulate clinical thinking, **40 fuzzy IF-THEN rules were developed**. The **Mamdani fuzzy inference method** was used to create the system, and the final output was thresholded at 0.5 to provide binary classification results (Fuzzy_Result).

A fuzzy risk score and a final classification were generated by applying the fuzzy classifier to **768 valid patient data**. To determine the accuracy of the system, the findings were compared with real diagnoses (Outcome). The system accurately classified **558 out of 768 cases**, with a final accuracy of **72.66%**. This outcome shows excellent performance for a rule-based, transparent fuzzy system that does not need data fitting or training.

This result shows that **fuzzy logic may be a strong, interpretable substitute for black-box machine learning models**, especially in applications related to healthcare where explainability and trust are essential. Despite its flaws, the system provides a useful, understandable method of decision assistance and catches key patterns of diabetes risk.

5. References

- Hasan, M. K., Azfar, A., Ammar, M., Islam, S. H., & Alazab, M. (2025). Secure biometric authentication for e-health systems using deep learning and blockchain technology. *Software Impacts*, 16, 100569. Retrieved from <https://www.sciencedirect.com/science/article/pii/S235271102500113X>
- Gupta, P., Agrawal, R., & Tiwari, V. N. (2020). A biometric authentication system for IoT devices using ECG signals. *2020 4th International Conference on Intelligent Computing and Control Systems (ICICCS)*, 119–124. Retrieved from <https://ieeexplore.ieee.org/abstract/document/9177565/>
- Banirostan, T., & Karimian, A. (2020). An intelligent multimodal biometric authentication system based on score level fusion using SVM. *International Journal of Computational Intelligence Systems*, 13(1), 1305–1314. Retrieved from <https://link.springer.com/article/10.2991/ijcis.d.201012.002>
- Alzahrani, B., & Alomar, N. (2023). Deep learning approach for biometric authentication using face and fingerprint modalities. *Machines*, 11(5), 572. Retrieved from <https://www.mdpi.com/2075-1702/11/5/572>
- Gairola, R., & Ganesan, R. (2024). A secured biometric-based authentication system for smart healthcare using machine learning. *IOP Conference Series: Materials Science and Engineering*, 1012(1), 012037. Retrieved from <https://iopscience.iop.org/article/10.1088/1757-899X/1012/1/012037/meta>

6. Source Code

Required Libraries

```
[ ] !pip install -q kagglehub scikit-fuzzy

import kagglehub
from kagglehub import KaggleDatasetAdapter
import pandas as pd
import numpy as np
import skfuzzy as fuzz
import skfuzzy.control as ctrl
import matplotlib.pyplot as plt
```

Load Dataset

```
[ ] df = kagglehub.load_dataset(
    KaggleDatasetAdapter.PANDAS,
    "akshaydattatraykhare/diabetes-dataset",
    "diabetes.csv"
)
```

Define Fuzzy Input Variables

```
[ ] pregnancies = ctrl.Antecedent(np.arange(0, 18, 1), 'Pregnancies')
glucose = ctrl.Antecedent(np.arange(0, 210, 1), 'Glucose')
blood_pressure = ctrl.Antecedent(np.arange(0, 140, 1), 'BloodPressure')
skin_thickness = ctrl.Antecedent(np.arange(0, 110, 1), 'SkinThickness')
insulin = ctrl.Antecedent(np.arange(0, 900, 1), 'Insulin')
bmi = ctrl.Antecedent(np.arange(0, 75, 1), 'BMI')
dpf = ctrl.Antecedent(np.arange(0, 2.6, 0.01), 'DPF')
age = ctrl.Antecedent(np.arange(20, 101, 1), 'Age')
```

Define Fuzzy Output Variables

```
[ ] risk = ctrl.Consequent(np.arange(0, 1.01, 0.01), 'Risk')
```

Pregnancies Membership Functions & Graph

```
[ ] pregnancies['None'] = fuzz.trapmf(pregnancies.universe, [0, 0, 1, 2])
pregnancies['Few'] = fuzz.trimf(pregnancies.universe, [1, 4, 7])
pregnancies['Many'] = fuzz.trapmf(pregnancies.universe, [5, 8, 17, 17])

plt.figure(figsize=(8, 4))
for term in pregnancies.terms:
    plt.plot(pregnancies.universe, pregnancies[term].mf, label=term)
plt.title("Pregnancies Membership Functions")
plt.xlabel("Pregnancies")
plt.ylabel("Membership Degree")
plt.ylim(0, 1.05)
plt.xlim(min(pregnancies.universe), max(pregnancies.universe))
plt.legend()
plt.grid(False)
plt.tight_layout()
plt.show()
```

Glucose Membership Function

```
[ ] glucose['Low'] = fuzz.trapmf(glucose.universe, [0, 0, 70, 100])
glucose['Medium'] = fuzz.trimf(glucose.universe, [90, 125, 160])
glucose['High'] = fuzz.trapmf(glucose.universe, [140, 170, 200, 210])

plt.figure(figsize=(8, 4))
for term in glucose.terms:
    plt.plot(glucose.universe, glucose[term].mf, label=term)
plt.title("Glucose Membership Functions")
plt.xlabel("Glucose (mg/dL)")
plt.ylabel("Membership Degree")
plt.ylim(0, 1.05)
plt.xlim(min(glucose.universe), max(glucose.universe))
plt.legend()
plt.grid(False)
plt.tight_layout()
plt.show()
```

Blood Pressure Membership Function

```
[ ] blood_pressure['Low'] = fuzz.trapmf(blood_pressure.universe, [0, 0, 50, 65])
blood_pressure['Normal'] = fuzz.trimf(blood_pressure.universe, [60, 75, 90])
blood_pressure['High'] = fuzz.trapmf(blood_pressure.universe, [85, 100, 130, 140])

plt.figure(figsize=(8, 4))
for term in blood_pressure.terms:
    plt.plot(blood_pressure.universe, blood_pressure[term].mf, label=term)
plt.title("Blood Pressure Membership Functions")
plt.xlabel("Blood Pressure (mm Hg)")
plt.ylabel("Membership Degree")
plt.ylim(0, 1.05)
plt.xlim(min(blood_pressure.universe), max(blood_pressure.universe))
plt.legend()
plt.grid(False)
plt.tight_layout()
plt.show()
```

Skin Thickness Membership Function

```
[ ] skin_thickness['Thin'] = fuzz.trapmf(skin_thickness.universe, [0, 0, 10, 20])
skin_thickness['Normal'] = fuzz.trimf(skin_thickness.universe, [15, 30, 45])
skin_thickness['Thick'] = fuzz.trapmf(skin_thickness.universe, [40, 55, 100, 110])

plt.figure(figsize=(8, 4))
for term in skin_thickness.terms:
    plt.plot(skin_thickness.universe, skin_thickness[term].mf, label=term)
plt.title("Skin Thickness Membership Functions")
plt.xlabel("Skin Thickness (mm)")
plt.ylabel("Membership Degree")
plt.ylim(0, 1.05)
plt.xlim(min(skin_thickness.universe), max(skin_thickness.universe))
plt.legend()
plt.grid(False)
plt.tight_layout()
plt.show()
```

Insulin Membership Function

```
[ ] insulin['Low'] = fuzz.trapmf(insulin.universe, [0, 0, 50, 100])
insulin['Medium'] = fuzz.trimf(insulin.universe, [80, 200, 320])
insulin['High'] = fuzz.trapmf(insulin.universe, [300, 500, 850, 900])

plt.figure(figsize=(8, 4))
for term in insulin.terms:
    plt.plot(insulin.universe, insulin[term].mf, label=term)
plt.title("Insulin Membership Functions")
plt.xlabel("Insulin (uU/mL)")
plt.ylabel("Membership Degree")
plt.ylim(0, 1.05)
plt.xlim(min(insulin.universe), max(insulin.universe))
plt.legend()
plt.grid(False)
plt.tight_layout()
plt.show()
```

BMI Membership Function

```
[ ] bmi['Low'] = fuzz.trapmf(bmi.universe, [0, 0, 18, 22])
bmi['Medium'] = fuzz.trimf(bmi.universe, [20, 27, 34])
bmi['High'] = fuzz.trapmf(bmi.universe, [30, 35, 70, 75])

plt.figure(figsize=(8, 4))
for term in bmi.terms:
    plt.plot(bmi.universe, bmi[term].mf, label=term)
plt.title("BMI Membership Functions")
plt.xlabel("BMI (kg/m²)")
plt.ylabel("Membership Degree")
plt.ylim(0, 1.05)
plt.xlim(min(bmi.universe), max(bmi.universe))
plt.legend()
plt.grid(False)
plt.tight_layout()
plt.show()
```

DPF Membership Function

```
[ ] dpf['Low'] = fuzz.trapmf(dpf.universe, [0.0, 0.0, 0.3, 0.5])
dpf['Medium'] = fuzz.trimf(dpf.universe, [0.4, 0.8, 1.2])
dpf['High'] = fuzz.trapmf(dpf.universe, [1.0, 1.5, 2.5, 2.6])

plt.figure(figsize=(8, 4))
for term in dpf.terms:
    plt.plot(dpf.universe, dpf[term].mf, label=term)
plt.title("DPF Membership Functions")
plt.xlabel("Diabetes Pedigree Function")
plt.ylabel("Membership Degree")
plt.ylim(0, 1.05)
plt.xlim(min(dpf.universe), max(dpf.universe))
plt.legend()
plt.grid(False)
plt.tight_layout()
plt.show()
```

Age Membership Function

```
[ ] age['Young'] = fuzz.trapmf(age.universe, [20, 20, 30, 40])
age['Middle'] = fuzz.trimf(age.universe, [35, 50, 65])
age['Old'] = fuzz.trapmf(age.universe, [60, 70, 100, 100])

plt.figure(figsize=(8, 4))
for term in age.terms:
    plt.plot(age.universe, age[term].mf, label=term)
plt.title("Age Membership Functions")
plt.xlabel("Age (years)")
plt.ylabel("Membership Degree")
plt.ylim(0, 1.05)
plt.xlim(min(age.universe), max(age.universe))
plt.legend()
plt.grid(False)
plt.tight_layout()
plt.show()
```

Risk Output

```
[ ] risk['Low'] = fuzz.trapmf(risk.universe, [0.0, 0.0, 0.2, 0.4])
risk['Medium'] = fuzz.trimf(risk.universe, [0.3, 0.5, 0.7])
risk['High'] = fuzz.trapmf(risk.universe, [0.6, 0.8, 1.0, 1.0])

for term in risk.terms:
    plt.plot(risk.universe, risk[term].mf, label=term)
    plt.fill_between(risk.universe, 0, risk[term].mf, alpha=0.2)
plt.title("Output Risk Membership Functions")
plt.xlabel("Risk Score")
plt.ylabel("Membership Degree")
plt.ylim(0, 1.05)
plt.xlim(min(risk.universe), max(risk.universe))
plt.legend()
plt.grid(False)
plt.tight_layout()
plt.show()
```

Define High Risk Rules

```
[ ] rule1 = ctrl.Rule(glucose['High'] & bmi['High'], risk['High'])
rule2 = ctrl.Rule(glucose['High'] & age['Old'], risk['High'])
rule3 = ctrl.Rule(glucose['High'] & insulin['High'], risk['High'])
rule4 = ctrl.Rule(glucose['High'] & dpf['High'], risk['High'])
rule5 = ctrl.Rule(pregnancies['Many'] & glucose['High'], risk['High'])
rule6 = ctrl.Rule(bmi['High'] & dpf['High'], risk['High'])
rule7 = ctrl.Rule(glucose['High'] & age['Middle'] & bmi['High'], risk['High'])
rule8 = ctrl.Rule(blood_pressure['High'] & bmi['High'], risk['High'])
rule9 = ctrl.Rule(insulin['High'] & age['Old'], risk['High'])
rule10 = ctrl.Rule(glucose['High'] & skin_thickness['Thick'] & insulin['High'], risk['High'])
rule11 = ctrl.Rule(glucose['High'] & bmi['High'] & pregnancies['Many'], risk['High'])
rule12 = ctrl.Rule(dpf['High'] & age['Old'], risk['High'])
rule13 = ctrl.Rule(bmi['High'] & skin_thickness['Thick'], risk['High'])
```

Define Medium Risk Rules

```
[ ] rule14 = ctrl.Rule(glucose['Medium'] & bmi['Medium'], risk['Medium'])
rule15 = ctrl.Rule(glucose['Medium'] & age['Middle'], risk['Medium'])
rule16 = ctrl.Rule(bmi['Medium'] & dpf['Medium'], risk['Medium'])
rule17 = ctrl.Rule(pregnancies['Few'] & glucose['Medium'], risk['Medium'])
rule18 = ctrl.Rule(insulin['Medium'] & glucose['Medium'], risk['Medium'])
rule19 = ctrl.Rule(blood_pressure['Normal'] & bmi['Medium'], risk['Medium'])
rule20 = ctrl.Rule(age['Middle'] & dpf['Medium'], risk['Medium'])
rule21 = ctrl.Rule(skin_thickness['Normal'] & bmi['Medium'], risk['Medium'])
rule22 = ctrl.Rule(blood_pressure['High'] & glucose['Medium'], risk['Medium'])
rule23 = ctrl.Rule(pregnancies['Many'] & age['Middle'], risk['Medium'])
rule24 = ctrl.Rule(glucose['Medium'] & dpf['High'], risk['Medium'])
rule25 = ctrl.Rule(insulin['Medium'] & bmi['Medium'], risk['Medium'])
rule26 = ctrl.Rule(glucose['High'] & dpf['Medium'], risk['Medium'])
rule27 = ctrl.Rule(skin_thickness['Thick'] & bmi['Medium'], risk['Medium'])
```

Define Low Risk Rules

```
[ ] rule28 = ctrl.Rule(glucose['Low'] & bmi['Low'], risk['Low'])
rule29 = ctrl.Rule(glucose['Low'] & age['Young'], risk['Low'])
rule30 = ctrl.Rule(glucose['Low'] & dpf['Low'], risk['Low'])
rule31 = ctrl.Rule(pregnancies['None'] & glucose['Low'], risk['Low'])
rule32 = ctrl.Rule(blood_pressure['Normal'] & glucose['Low'], risk['Low'])
rule33 = ctrl.Rule(bmi['Low'] & skin_thickness['Thin'], risk['Low'])
rule34 = ctrl.Rule(insulin['Low'] & glucose['Low'], risk['Low'])
rule35 = ctrl.Rule(dpf['Low'] & age['Young'], risk['Low'])
rule36 = ctrl.Rule(insulin['Low'] & bmi['Low'], risk['Low'])
rule37 = ctrl.Rule(pregnancies['None'] & dpf['Low'] & glucose['Low'], risk['Low'])
rule38 = ctrl.Rule(age['Young'] & bmi['Low'], risk['Low'])
rule39 = ctrl.Rule(glucose['Low'] & skin_thickness['Thin'], risk['Low'])
rule40 = ctrl.Rule(blood_pressure['Low'] & glucose['Low'], risk['Low'])
```

Combine All Rules Into a List

```
[ ] rules = [
    rule1, rule2, rule3, rule4, rule5, rule6, rule7, rule8, rule9, rule10,
    rule11, rule12, rule13, rule14, rule15, rule16, rule17, rule18, rule19, rule20,
    rule21, rule22, rule23, rule24, rule25, rule26, rule27,
    rule28, rule29, rule30, rule31, rule32, rule33, rule34, rule35, rule36, rule37, rule38, rule39, rule40
]
```

Create Control System and Simulation Engine

```
[ ] risk_ctrl = ctrl.ControlSystem(rules)
risk_simulation = ctrl.ControlSystemSimulation(risk_ctrl)
```

Run Simulation on all Rows

```
[ ] risk_scores = []
    fuzzy_results = []

    for i, row in df.iterrows():
        try:
            # Feed inputs into the fuzzy system
            risk_simulation.input['Pregnancies'] = row['Pregnancies']
            risk_simulation.input['Glucose'] = row['Glucose']
            risk_simulation.input['BloodPressure'] = row['BloodPressure']
            risk_simulation.input['SkinThickness'] = row['SkinThickness']
            risk_simulation.input['Insulin'] = row['Insulin']
            risk_simulation.input['BMI'] = row['BMI']
            risk_simulation.input['DPF'] = row['DiabetesPedigreeFunction']
            risk_simulation.input['Age'] = row['Age']

            # Compute output
            risk_simulation.compute()
            score = risk_simulation.output['Risk']
            risk_scores.append(score)

            # Apply threshold for binary class
            fuzzy_results.append(1 if score >= 0.5 else 0)

        except:
            # Handle fuzzy system errors (e.g. values outside MF range)
            risk_scores.append(None)
            fuzzy_results.append(None)

    # Store results in dataframe
    df['Risk_Score'] = risk_scores
    df['Fuzzy_Result'] = fuzzy_results
    df[['Risk_Score', 'Fuzzy_Result']].head()

    # Optional: Create labeled result (for clarity in the dataset)
    df_cleaned = df.dropna(subset=['Fuzzy_Result']).copy()
    df_cleaned['Fuzzy_Classification'] = df_cleaned['Fuzzy_Result'].replace({1: 'Diabetic', 0: 'Non-Diabetic'})

    pd.set_option('display.max_columns', None) # Show all columns
    pd.set_option('display.width', None)      # Auto-resize to fit width
    df_cleaned

    df_cleaned.to_csv('final_fuzzy_dataset.csv', index=False)
```

Evaluate Classification Accuracy

```
[ ] total_predictions = len(df_cleaned)
    correct_predictions = (df_cleaned['Fuzzy_Result'] == df_cleaned['Outcome']).sum()
    accuracy = (correct_predictions / total_predictions) * 100
```

Output Summary

```
[ ] print(f"Total Predictions: {total_predictions}")
    print(f"Correct Predictions: {correct_predictions}")
    print(f"Accuracy: {accuracy:.2f}%")
```

7. Presentation Link

<https://youtu.be/5I6ihBsRfiw>